

Ex. No:6 Use Sleuth Kit to analyze digital evidence

Description

The Sleuth Kit (TSK) is a collection of command-line tools that allow you to analyze disk images and recover digital evidence. Here's a step-by-step guide to using Sleuth Kit on a Windows machine to analyze digital evidence:

Step 1: Install Sleuth Kit

1. Download Sleuth Kit:

- Visit the official [Sleuth Kit website](https://www.sleuthkit.org/) or https://drive.google.com/drive/u/1/folders/1ilSFY7Tqn2L7AjQGhq8yJ8kixc_xTU-v
- and download the latest version for Windows.

2. Install Sleuth Kit:

- Run the installer and follow the instructions to install Sleuth Kit on your Windows machine.

Step 2: Acquire the Disk Image

Before analysis, you need a disk image of the evidence. This can be an image of a hard drive, memory card, or any other storage device.

3. Create Disk Image:

- Use a tool like FTK Imager or dd to create a bit-by-bit copy of the storage device. Ensure the image is in a format supported by Sleuth Kit, such as .dd, .raw, .img, or .E01.
- Download the below files from the google drive
 1. 4Dell Latitude CPi.E01
 2. 4Dell Latitude CPi.E02

Step 3: Mount the Disk Image (Optional)

Mounting the disk image makes it easier to analyze the file system.

4. Mount the Image:

- You can use a tool like OSFMount to mount the image as a virtual drive on your Windows system.
- This step is optional but helps with navigating the file system easily.

Step 4: Analyze the File System

Use Sleuth Kit tools to analyze the file system and locate evidence.

5. Navigate to the Sleuth Kit Directory:

- Open the Command Prompt and navigate to the directory where Sleuth Kit is installed.

6. Identify File System Type with fsstat:

- Run the command:

arduino

```
fsstat [image file] > filesystem_info.txt
```

- This command outputs information about the file system, which is crucial for understanding the structure of the disk.

3. List Partitions with mmls:

- Run the command:

arduino

```
mmls [image file] > partitions.txt
```

- This command lists the partitions within the image file.

4. Analyze File System with fls:

- Run the command:

arduino

```
fls -r [image file] > file_list.txt
```

- This command recursively lists files and directories in the file system, showing their metadata.

5. Recover Deleted Files with icat:

- To extract a specific file, use icat:

CSS

```
icat [image file] [inode number] > [output file]
```

- Replace [inode number] with the inode of the file you want to recover, which you can find from the fls output.

Step 5: Analyze Metadata

Extract metadata from files to understand more about the file's history.

6. View Metadata with istat:

- Run the command:

CSS

```
istat [image file] [inode number] > metadata_info.txt
```

- This provides detailed information about a file, including timestamps, size, and allocation status.

Step 6: Timeline Analysis (Optional)

Creating a timeline of file activity can be crucial in an investigation.

7. Create Timeline with mactime:

- Generate a body file using fls:

CSS

```
fls -m / -r [image file] > body.txt
```

- Then create the timeline:

CSS

mactime -b body.txt > timeline.txt

- o The timeline includes MAC (Modified, Accessed, Changed) times of files.

Step 7: Generate a Report

Document your findings by compiling all the data into a comprehensive report.

1. Compile the Data:

- o Gather all the output files (e.g., filesystem_info.txt, partitions.txt, file_list.txt, metadata_info.txt, timeline.txt).

2. Analyze and Document:

- o Review the findings, highlight important evidence, and write a report summarizing the investigation's results.

Step 8: Finalize and Store Evidence

Ensure that all evidence and reports are securely stored.

3. Archive Evidence:

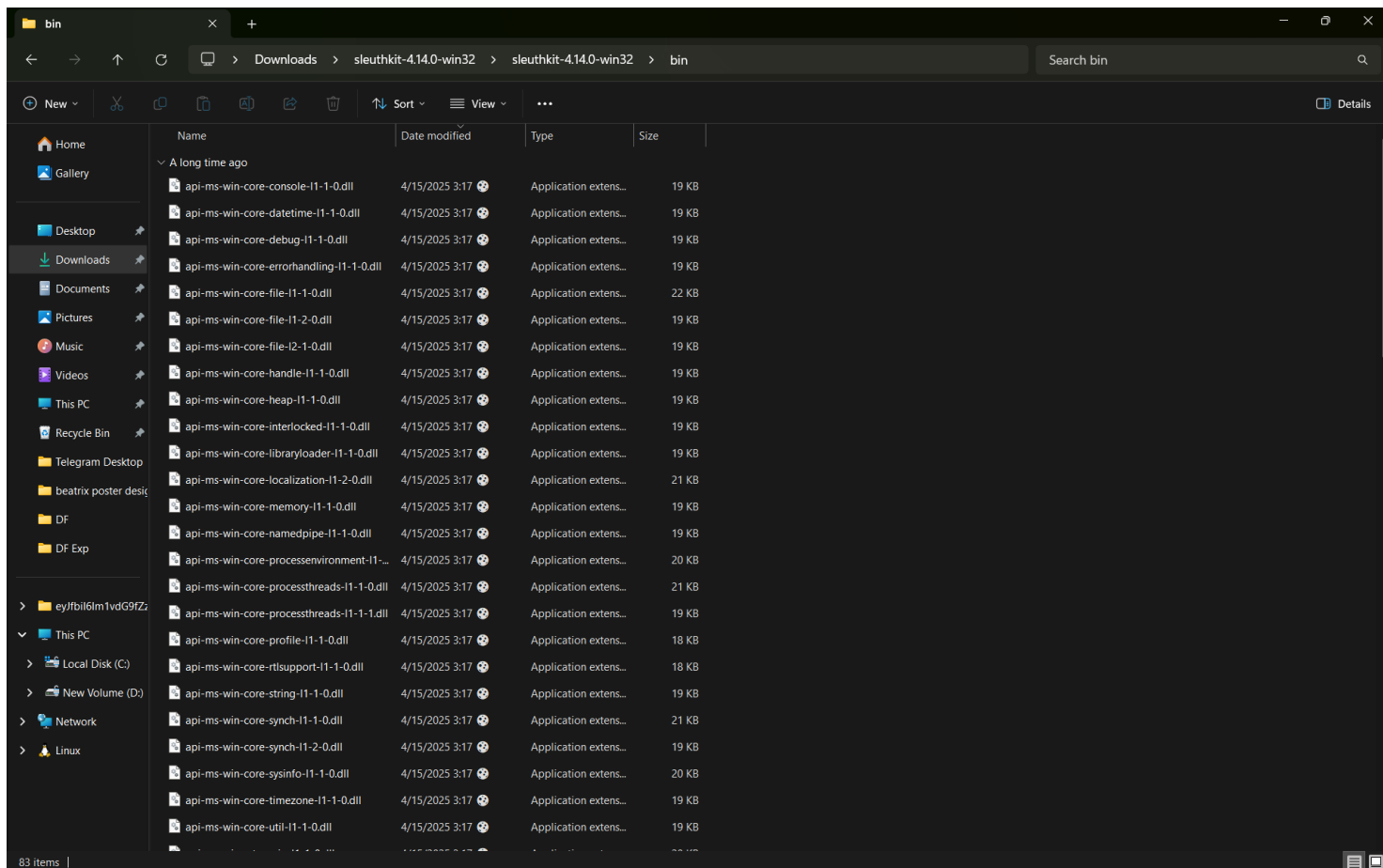
- o Use a secure method to archive the disk image and analysis results, ensuring integrity and confidentiality.

4. Store Securely:

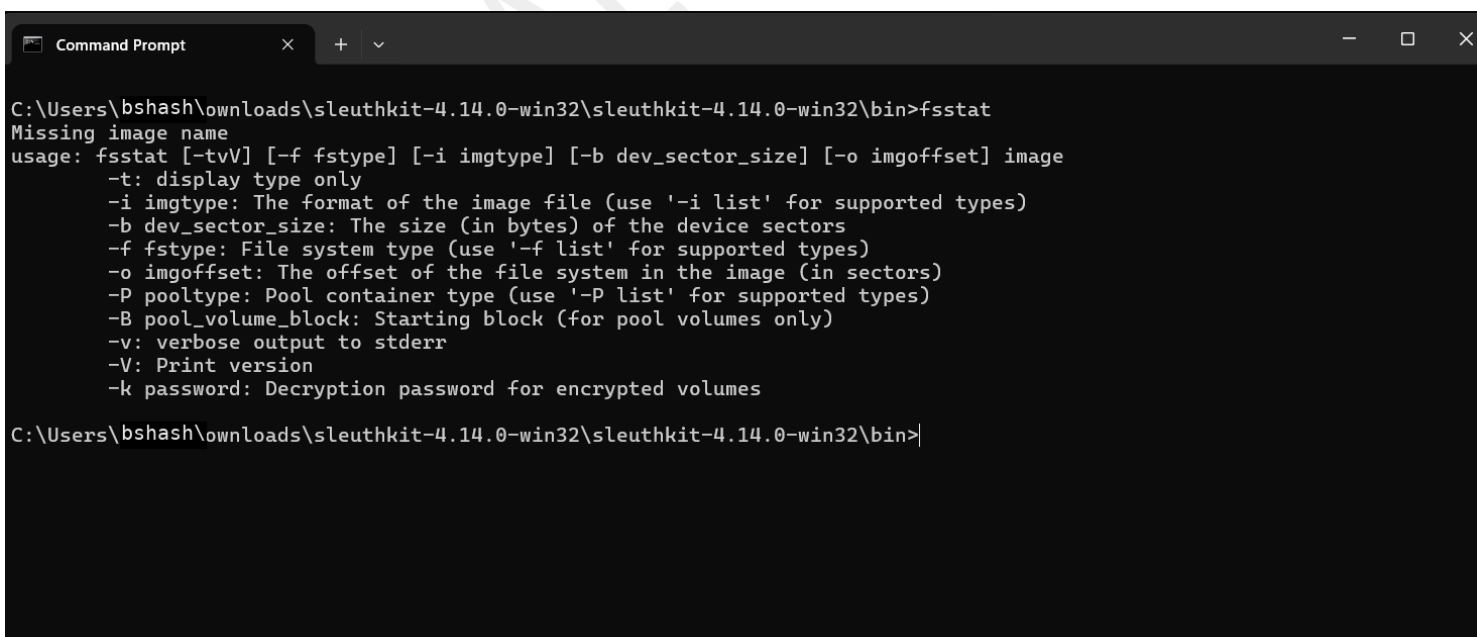
- o Store the archived data in a secure location, following the chain of custody procedures.

By following these steps, you can use Sleuth Kit on a Windows machine to effectively analyze digital evidence and extract crucial information for your investigation.

Fsstat command execution:



Identifying File System:



Installed Sleuth Kit:

```
C:\Users\bshash\downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fsstat -i ewf -o 63 "C:\Users\bshash\downloads\sleuthkit-4.14.0-win32\4Dell Latitude CPi (1).E01"
FILE SYSTEM INFORMATION
-----
File System Type: NTFS
Volume Serial Number: B26CB1CE6CB18D9B
OEM Name: NTFS
Version: Windows XP

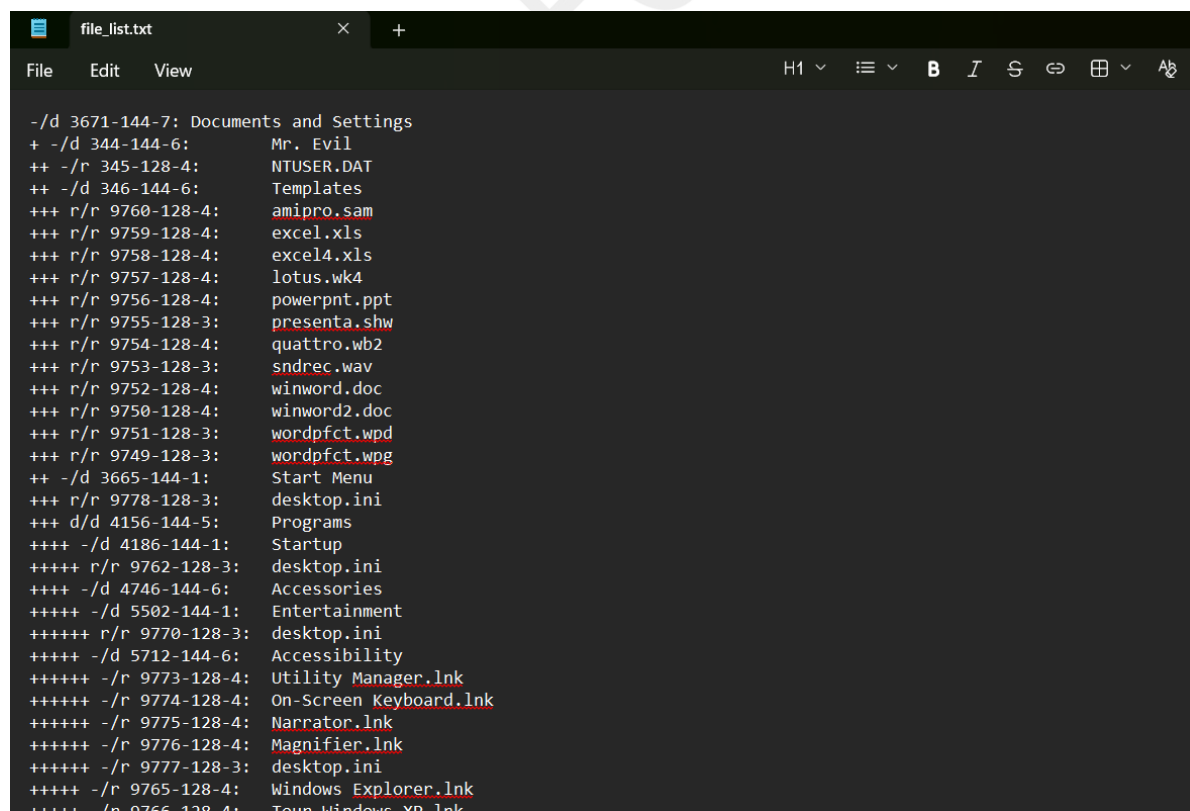
METADATA INFORMATION
-----
First Cluster of MFT: 2097152
First Cluster of MFT Mirror: 4755208
Size of MFT Entries: 1024 bytes
Size of Index Records: 4096 bytes
Range: 0 - 12305
Root Directory: 5

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 512
Total Cluster Range: 0 - 9510415
Total Sector Range: 0 - 9510415

$AttrDef Attribute Values:
$STANDARD_INFORMATION (16) Size: 48-72 Flags: Resident
$ATTRIBUTE_LIST (32) Size: No Limit Flags: Non-resident
$FILE_NAME (48) Size: 68-578 Flags: Resident,Index
$OBJECT_ID (64) Size: 0-256 Flags: Resident
$SECURITY_DESCRIPTOR (80) Size: No Limit Flags: Non-resident
$VOLUME_NAME (96) Size: 2-256 Flags: Resident
$VOLUME_INFORMATION (112) Size: 12-12 Flags: Resident
$DATA (128) Size: No Limit Flags:
```

Identifying File System:

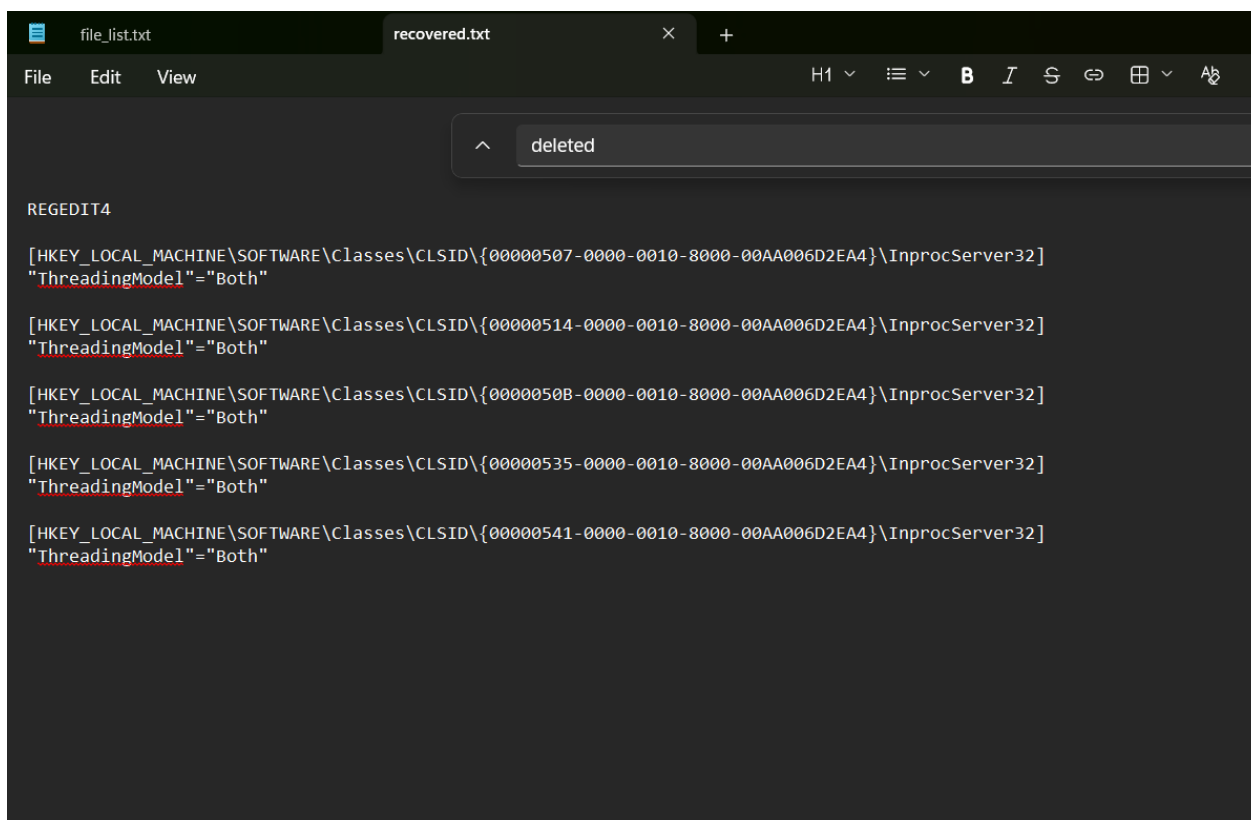
```
C:\Users\bshash\downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fls -i ewf -o 63 -r "C:\Users\bshash\downloads\sleuthkit-4.14.0-win32\4Dell Latitude CPi (1).E01" > file_list.txt
```



```
-/d 3671-144-7: Documents and Settings
+ -/d 344-144-6: Mr. Evil
++ -/r 345-128-4: NTUSER.DAT
++ -/d 346-144-6: Templates
+++ r/r 9760-128-4: amipro.sam
+++ r/r 9759-128-4: excel.xls
+++ r/r 9758-128-4: excel4.xls
+++ r/r 9757-128-4: lotus.wk4
+++ r/r 9756-128-4: powerpnt.ppt
+++ r/r 9755-128-3: presenta.shw
+++ r/r 9754-128-4: quattro.wb2
+++ r/r 9753-128-3: sndrec.wav
+++ r/r 9752-128-4: winword.doc
+++ r/r 9750-128-4: winword2.doc
+++ r/r 9751-128-3: wordpfct.wpd
+++ r/r 9749-128-3: wordpfct.wpg
++ -/d 3665-144-1: Start Menu
+++ r/r 9778-128-3: desktop.ini
+++ d/d 4156-144-5: Programs
++++ -/d 4186-144-1: Startup
+++++ r/r 9762-128-3: desktop.ini
+++++ -/d 4746-144-6: Accessories
+++++ -/d 5502-144-1: Entertainment
+++++ r/r 9770-128-3: desktop.ini
+++++ -/d 5712-144-6: Accessibility
+++++ -/r 9773-128-4: Utility Manager.lnk
+++++ -/r 9774-128-4: On-Screen Keyboard.lnk
+++++ -/r 9775-128-4: Narrator.lnk
+++++ -/r 9776-128-4: Magnifier.lnk
+++++ -/r 9777-128-3: desktop.ini
+++++ -/r 9765-128-4: Windows Explorer.lnk
+++++ -/r 9766-128-4: Tour Windows XP.lnk
```

Recovering Deleted File:

```
C:\Users\bshash\ownloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>icat -i ewf -o 63 "C:\Users\bshash\ownloads\sleuthkit-4.14.0-win32\4Dell Latitude CPi (1).E01" 150 > recovered.txt
```



Metadata:

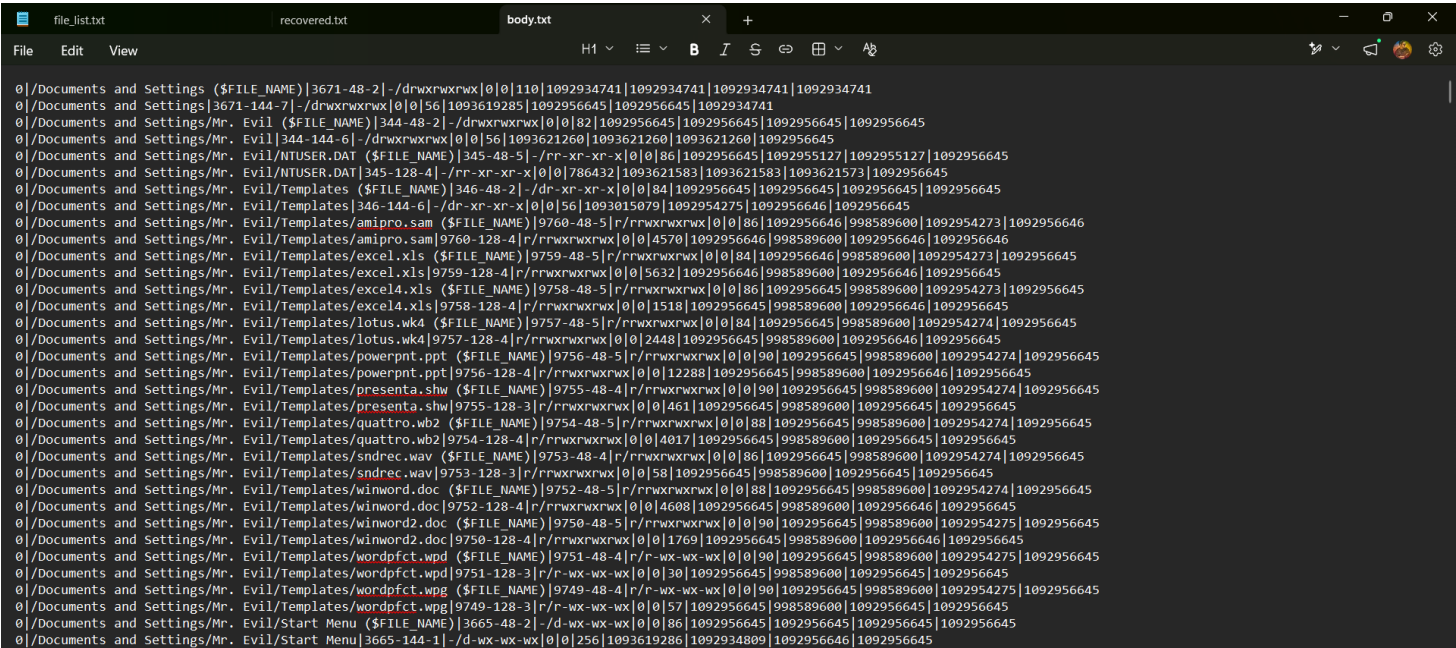
```
C:\Users\bshash\ownloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>istat -i ewf -o 63 "C:\Users\bshash\ownloads\sleuthkit-4.14.0-win32\4Dell Latitude CPi (1).E01" 150
MFT Entry Header Values:
Entry: 150      Sequence: 150
$LogFile Sequence Number: 9239131
Allocated File
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags: Archive
Owner ID: 0
Security ID: 0 ()
Version 16 of 16943392
Created: 1999-04-24 03:52:00.000000000 (India Standard Time)
File Modified: 1999-04-24 03:52:00.000000000 (India Standard Time)
MFT Modified: 2004-08-19 22:32:11.098512000 (India Standard Time)
Accessed: 2004-08-18 05:30:00.000000000 (India Standard Time)

$FILE_NAME Attribute Values:
Flags: Archive
Name: ADOFRE15.REG
Parent MFT Entry: 137 Sequence: 137
Allocated Size: 1024 Actual Size: 640
Created: 1999-04-24 03:52:00.000000000 (India Standard Time)
File Modified: 1999-04-24 03:52:00.000000000 (India Standard Time)
MFT Modified: 1999-04-24 03:52:00.000000000 (India Standard Time)
Accessed: 2004-08-18 05:30:00.000000000 (India Standard Time)
```

Generating Timeline:

```
C:\Users\bshash\ownloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fls -i ewf
-o 63 -m / -r "C:\Users\bshash\ownloads\sleuthkit-4.14.0-win32\4Dell Latitude CPi (1).
E01" > body.txt
```



Gathering all the generated Files(Report):

